

BFS and A* 8-Puzzle Comparison

Initial State	Goal State
8 7 6	1 2 3
5 4 3	4 5 6
2 1 0	7 8 0

Measured Results

Measurement	BFS	A*
Optimal moves	30	30
States expanded	181,393	7,330
States discovered	181,440	10,895
Priority-queue entries generated	—	11,285

Explanation

Both algorithms find an optimal solution containing **30 moves**, but A* expands far fewer states than BFS.

Breadth-First Search (BFS) explores states according to their distance from the start. It expands every state at one search depth before moving to the next depth. Because BFS has no estimate of which states are closer to the goal, it examines many states that do not lead toward the solution efficiently.

A* explores states according to $f(n) = g(n) + h(n)$. Here, $g(n)$ is the number of moves from the initial state to the current state, while $h(n)$ estimates the number of moves remaining.

This implementation uses **Manhattan distance** for $h(n)$. For every numbered tile, Manhattan distance counts the horizontal and vertical moves needed to reach its goal position. The blank tile is not included.

Manhattan distance guides A* toward states whose tiles are closer to their goal locations. It never overestimates the true remaining cost, because each tile must move at least that many grid spaces. Therefore, A* still guarantees an optimal solution while avoiding most of the unpromising states explored by BFS.